Md. Rasel Khondokar

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Ph.D. student and machine learning engineer specializing in NLP, retrieval, and scalable ML systems, with 4+ years of industry experience in production ML, distributed training, and cloud deployment. Seeking a Summer 2027 ML internship.

PUBLICATIONS

- Bridging Lexical Divergence: LLM-Assisted, Cost-Efficient, Zero-Shot Scientific Entity Linking. *Findings of EMNLP*, 2026.
- Enhancing Medical AI Safety Through Multi-Agent Evaluation and Iterative Refinement. *ACM BCB*, 2025.
- Improving the Safety and Trustworthiness of Medical AI via Multi-Agent Evaluation Loops. Under review, 2026.

RESEARCH EXPERIENCE

Ph.D. Researcher – Data and Knowledge Mining Lab Jan 2024 – Present
Iowa State University

- Developed a zero-shot scientific entity-linking framework combining LLM-generated training data, retriever–reranker fine-tuning, and ontology-aware semantic filtering; evaluated on NCBI Disease and BC5CDR.
- Built and annotated an animal-science dataset linking quantitative trait locus (QTL) mentions to standardized ontologies.
- Investigate knowledge graphs and retrieval-augmented generation (RAG) for scientific knowledge extraction and question answering.
- Researched multi-agent evaluation and iterative refinement to improve the safety and trustworthiness of medical AI outputs.

PROFESSIONAL EXPERIENCE

Lead Software Engineer – Machine Learning/NLP Jan 2022 – Dec 2023
Technometrics Limited, Dhaka, Bangladesh

- Architected a multi-node ML inference pipeline using Celery queues and per-task GPU allocation to serve automatic speech recognition (ASR), speaker diarization, and text classification models.
- Developed a BERT-based Bengali topic classifier with 85% accuracy across 17 categories using PyTorch and Distributed Data Parallel (DDP) training; built sentiment and cyberbullying models with 84% and 81% accuracy.
- Built a hybrid transformer and rule-based punctuation restoration system with 75% accuracy on noisy ASR transcripts.
- Developed aspect-based sentiment analysis with 80% aspect extraction and 72% sentiment accuracy; authored annotation guidelines and managed the labeling pipeline.

Software Engineer – Machine Learning Jun 2019 – Dec 2021
Ishraak Solutions Limited, Dhaka, Bangladesh

- Led a five-member ML team developing production systems serving thousands of daily users.
- Built an NLP information-extraction system that reduced manual processing time by 70%; developed a BERT and spaCy named entity recognition model achieving 73% F1.
- Developed a job categorization model achieving 81% accuracy across 50 categories and a hybrid job recommender combining collaborative and content-based filtering.
- Built conversational AI applications using Dialogflow and Rasa.

TECHNICAL SKILLS

Programming & Tools: Python, SQL, Java, Git, Linux, FastAPI, Flask
ML Frameworks: PyTorch, Hugging Face Transformers, TensorFlow, Keras, NeMo
NLP & Retrieval: LLMs, entity linking, RAG, knowledge graphs, spaCy, LangChain, LlamaIndex
ML Infrastructure: DDP, Docker, Kubernetes, AWS SageMaker, MLflow, TorchServe, Celery, Redis
Data Systems: MongoDB, Elasticsearch, Neo4j, Pinecone, Weaviate

EDUCATION

Ph.D. in Computer Science Jan 2024 – Present
Iowa State University, Ames, IA GPA: 3.79/4.00
M.S. in Computer Science 2026
Iowa State University, Ames, IA GPA: 3.76/4.00
B.Sc. in Computer Science and Engineering 2019
Daffodil International University, Bangladesh CGPA: 3.80/4.00

CERTIFICATIONS

AWS Certified Machine Learning – Specialty | NLP with Classification and Vector Spaces | Docker Essentials